

A proof of the conjectured recurrence for OEIS A000483

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Abstract

OEIS A000483 carries a conjectured linear recurrence with polynomial coefficients, of order 3. We prove it. The entry's exponential generating function is transcendental, so the usual algebraic arguments do not apply; instead we observe that it lies in a finitely generated module over $\mathbb{Q}(x)$ spanned by monomials $\log(u)^a \exp(g)^k$, and that this module is closed under differentiation, because $(\log u)' = u'/u$ and $(\exp g)' = g' \exp g$ lower the log power and fix the exponential respectively. Re-indexing the recurrence forward turns shifts into derivatives, so the sequence $b(m) = \sum_j q_j(m) a(m+j)$ has exponential generating function $B = \sum_j q_j(\theta) A^{(j)}$, computed exactly inside that module. Here B collapses to the polynomial $20x^3$, of degree 3, which proves the recurrence for every $n > 3$.

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1 The sequence and the conjecture

OEIS A000483 is “Associated Stirling numbers: second-order reciprocal Stirling numbers (Fekete) $a(n) = [[n, 3]]$. The number of 3-orbit permutations of an n -set with at least 2 elements in each orbit”. It has offset 6 and begins

$$a(6), \dots, a(13) = 15, 210, 2380, 26432, 303660, 3678840, 47324376, 647536032, \dots$$

The entry gives the exponential generating function

$$A(x) = \sum_{n \geq 6} a(n) \frac{x^n}{n!} = -\frac{(x + \log(1-x))^3}{6}, \quad (1)$$

and it carries the following comment:

Conjecture: $(n-2)^*(n-4)*a(n) - (n-1)^*(3*n^2-21*n+35)*a(n-1) + (n-1)^*(n-2)^*(3*n^2-24*n+47)*a(n-2) - (n-5)^*(n-1)^*(n-2)^*(n-3)^*(n-4)*a(n-3) = 0$.

As of the “Last modified” line on the live entry (2026-07-11, revision 42) the statement is still recorded as a conjecture, and no proof appears among the entry's links, formulas or programs.

Writing the conjectured relation in the form $\sum_{i=0}^3 p_i(n) a(n-i) = 0$, the coefficient polynomials are

$$p_0(n) = (n-4)(n-2), \quad p_1(n) = -(n-1)(3n^2-21n+35), \quad p_2(n) = (n-2)(n-1)(3n^2-24n+47),$$

2 Recurrences as differential operators

Let $A(x) = \sum_{n \geq 0} a(n)x^n/n!$ be the exponential generating function of a sequence, and let $\theta = x \frac{d}{dx}$, so that θ acts on x^m as multiplication by m . For a polynomial p we write $p(\theta)$ for the corresponding differential operator, so that $p(\theta)A = \sum_m p(m)a(m)x^m/m!$.

Lemma 1. Let $p_0, \dots, p_r$ be polynomials and suppose the conjecture asserts $\sum_{i=0}^r p_i(n) a(n-i) = 0$. Re-index by $n = m + r$ and $j = r - i$, and write $q_j(m) = p_{r-j}(m + r)$, so the claim reads $\sum_{j=0}^r q_j(m) a(m + j) = 0$. Put $b(m) = \sum_j q_j(m) a(m + j)$. Then

$$B(x) := \sum_{m \geq 0} b(m) \frac{x^m}{m!} = \sum_{j=0}^r (q_j(\theta) A^{(j)})(x),$$

where $A^{(j)}$ denotes the j -th derivative of A .

Proof. For an exponential generating function an upward shift is differentiation: $\sum_{m \geq 0} a(m + j) x^m / m! = A^{(j)}(x)$. Multiplying the coefficient of $x^m / m!$ by $q_j(m)$ is exactly the action of $q_j(\theta)$, and summing over j gives the statement. The re-indexing is what makes this work: a downward shift $a(n - i)$ would correspond to repeated integration, whereas an upward shift is a derivative, and the field used below is closed under differentiation. $\square$

Corollary 2. With the notation of Lemma 1, the recurrence $\sum_i p_i(n) a(n - i) = 0$ holds for every $n > d + r$ if and only if B is a polynomial of degree at most d .

Proof. $b(m)$ is $m!$ times the coefficient of x^m in B , so $b(m) = 0$ for all $m > d$ says exactly that B has no terms above x^d . $\square$

This turns the conjecture into a closed-form identity. The point is that it can then be decided exactly, with no truncation: A is algebraic, so B lies in the same field as A , and one only has to recognise it.

Lemma 3. Let $u, g \in \mathbb{Q}(x)$ and let M be the $\mathbb{Q}(x)$ -module spanned by the monomials $\log(u)^a \exp(g)^k$ for $a \geq 0$ and $k \in \mathbb{Z}$. Then M is closed under d/dx , and hence under $\theta = x d/dx$; explicitly,

$$\frac{d}{dx} \left(c \log(u)^a \exp(g)^k \right) = \left(c' + c k g' \right) \log(u)^a \exp(g)^k + c a \frac{u'}{u} \log(u)^{a-1} \exp(g)^k, \quad c \in \mathbb{Q}(x).$$

Proof. Differentiate the product. The factor $\exp(g)^k = \exp(kg)$ contributes kg' and does not change the monomial; the factor $\log(u)^a$ contributes $a(u'/u) \log(u)^{a-1}$, lowering the log exponent by one. Both new coefficients lie in $\mathbb{Q}(x)$, so the result is again a finite combination of the same monomials. $\square$

Two features make this usable. The exponential part never mixes monomials at all, and the logarithmic part only ever moves *downwards*, so the module generated by any single element is finite dimensional and every computation terminates. Writing B in this basis, the test of Corollary 2 becomes: B is a polynomial precisely when every coefficient outside the monomial $\log(u)^0 \exp(g)^0$ vanishes and the remaining coefficient has constant denominator. Both are decided by exact cancellation of rational functions, so no series truncation enters anywhere.

3 The computation for A000483

The generating function (1) lies in the module M of Lemma 3. Applying Lemma 1 to the coefficient polynomials of Section 1 and reducing in K by Lemma 3, the $\sqrt{D}$ component of B cancels identically and the rational part collapses to

$$B(x) = 20x^3,$$

a polynomial of degree 3.

Theorem 4. *The conjectured recurrence*

$(n-4)(n-2)a(n)-(n-1)(3n^2-21n+35)a(n-1)+(n-2)(n-1)(3n^2-24n+47)a(n-2)-(n-5)(n-4)a(n-3)+$
 $(n-1)(n-2)(n-3)(3n^2-21n+35)a(n-4)+$
 $(n-2)(n-1)(n-3)(n-4)(3n^2-24n+47)a(n-5)+$
 $(n-3)(n-2)(n-4)(n-5)(3n^2-21n+35)a(n-6)+$
 $(n-4)(n-3)(n-5)(n-6)(3n^2-24n+47)a(n-7)+$
 $(n-5)(n-4)(n-6)(n-7)(3n^2-21n+35)a(n-8)+$
 $(n-6)(n-5)(n-7)(n-8)(3n^2-24n+47)a(n-9)+$
 $(n-7)(n-6)(n-8)(n-9)(3n^2-21n+35)a(n-10)+$
 $(n-8)(n-7)(n-9)(n-10)(3n^2-24n+47)a(n-11)+$
 $(n-9)(n-8)(n-10)(n-11)(3n^2-21n+35)a(n-12)+$
 $(n-10)(n-9)(n-11)(n-12)(3n^2-24n+47)a(n-13)+$
 $(n-11)(n-10)(n-12)(n-13)(3n^2-21n+35)a(n-14)+$
 $(n-12)(n-11)(n-13)(n-14)(3n^2-24n+47)a(n-15)+$
 $(n-13)(n-12)(n-14)(n-15)(3n^2-21n+35)a(n-16)+$
 $(n-14)(n-13)(n-15)(n-16)(3n^2-24n+47)a(n-17)+$
 $(n-15)(n-14)(n-16)(n-17)(3n^2-21n+35)a(n-18)+$
 $(n-16)(n-15)(n-17)(n-18)(3n^2-24n+47)a(n-19)+$
 $(n-17)(n-16)(n-18)(n-19)(3n^2-21n+35)a(n-20)+$
 $(n-18)(n-17)(n-19)(n-20)(3n^2-24n+47)a(n-21)+$
 $(n-19)(n-18)(n-20)(n-21)(3n^2-21n+35)a(n-22)+$
 $(n-20)(n-19)(n-21)(n-22)(3n^2-24n+47)a(n-23)+$
 $(n-21)(n-20)(n-22)(n-23)(3n^2-21n+35)a(n-24)+$
 $(n-22)(n-21)(n-23)(n-24)(3n^2-24n+47)a(n-25)+$
 $(n-23)(n-22)(n-24)(n-25)(3n^2-21n+35)a(n-26)+$
 $(n-24)(n-23)(n-25)(n-26)(3n^2-24n+47)a(n-27)+$
 $(n-25)(n-24)(n-26)(n-27)(3n^2-21n+35)a(n-28)+$
 $(n-26)(n-25)(n-27)(n-28)(3n^2-24n+47)a(n-29)+$
 $(n-27)(n-26)(n-28)(n-29)(3n^2-21n+35)a(n-30)+$
 $(n-28)(n-27)(n-29)(n-30)(3n^2-24n+47)a(n-31)+$
 $(n-29)(n-28)(n-30)(n-31)(3n^2-21n+35)a(n-32)+$
 $(n-30)(n-29)(n-31)(n-32)(3n^2-24n+47)a(n-33)+$
 $(n-31)(n-30)(n-32)(n-33)(3n^2-21n+35)a(n-34)+$
 $(n-32)(n-31)(n-33)(n-34)(3n^2-24n+47)a(n-35)+$
 $(n-33)(n-32)(n-34)(n-35)(3n^2-21n+35)a(n-36)+$
 $(n-34)(n-33)(n-35)(n-36)(3n^2-24n+47)a(n-37)+$
 $(n-35)(n-34)(n-36)(n-37)(3n^2-21n+35)a(n-38)+$
 $(n-36)(n-35)(n-37)(n-38)(3n^2-24n+47)a(n-39)+$
 $(n-37)(n-36)(n-38)(n-39)(3n^2-21n+35)a(n-40)+$
 $(n-38)(n-37)(n-39)(n-40)(3n^2-24n+47)a(n-41)+$
 $(n-39)(n-38)(n-40)(n-41)(3n^2-21n+35)a(n-42)+$
 $(n-40)(n-39)(n-41)(n-42)(3n^2-24n+47)a(n-43)+$
 $(n-41)(n-40)(n-42)(n-43)(3n^2-21n+35)a(n-44)+$
 $(n-42)(n-41)(n-43)(n-44)(3n^2-24n+47)a(n-45)+$
 $(n-43)(n-42)(n-44)(n-45)(3n^2-21n+35)a(n-46)+$
 $(n-44)(n-43)(n-45)(n-46)(3n^2-24n+47)a(n-47)+$
 $(n-45)(n-44)(n-46)(n-47)(3n^2-21n+35)a(n-48)+$
 $(n-46)(n-45)(n-47)(n-48)(3n^2-24n+47)a(n-49)+$
 $(n-47)(n-46)(n-48)(n-49)(3n^2-21n+35)a(n-50)+$
 $(n-48)(n-47)(n-49)(n-50)(3n^2-24n+47)a(n-51)+$
 $(n-49)(n-48)(n-50)(n-51)(3n^2-21n+35)a(n-52)+$
 $(n-50)(n-49)(n-51)(n-52)(3n^2-24n+47)a(n-53)+$
 $(n-51)(n-50)(n-52)(n-53)(3n^2-21n+35)a(n-54)+$
 $(n-52)(n-51)(n-53)(n-54)(3n^2-24n+47)a(n-55)+$
 $(n-53)(n-52)(n-54)(n-55)(3n^2-21n+35)a(n-56)+$
 $(n-54)(n-53)(n-55)(n-56)(3n^2-24n+47)a(n-57)+$
 $(n-55)(n-54)(n-56)(n-57)(3n^2-21n+35)a(n-58)+$
 $(n-56)(n-55)(n-57)(n-58)(3n^2-24n+47)a(n-59)+$
 $(n-57)(n-56)(n-58)(n-59)(3n^2-21n+35)a(n-60)+$
 $(n-58)(n-57)(n-59)(n-60)(3n^2-24n+47)a(n-61)+$
 $(n-59)(n-58)(n-60)(n-61)(3n^2-21n+35)a(n-62)+$
 $(n-60)(n-59)(n-61)(n-62)(3n^2-24n+47)a(n-63)+$
 $(n-61)(n-60)(n-62)(n-63)(3n^2-21n+35)a(n-64)+$
 $(n-62)(n-61)(n-63)(n-64)(3n^2-24n+47)a(n-65)+$
 $(n-63)(n-62)(n-64)(n-65)(3n^2-21n+35)a(n-66)+$
 $(n-64)(n-63)(n-65)(n-66)(3n^2-24n+47)a(n-67)+$
 $(n-65)(n-64)(n-66)(n-67)(3n^2-21n+35)a(n-68)+$
 $(n-66)(n-65)(n-67)(n-68)(3n^2-24n+47)a(n-69)+$
 $(n-67)(n-66)(n-68)(n-69)(3n^2-21n+35)a(n-70)+$
 $(n-68)(n-67)(n-69)(n-70)(3n^2-24n+47)a(n-71)+$
 $(n-69)(n-68)(n-70)(n-71)(3n^2-21n+35)a(n-72)+$
 $(n-70)(n-69)(n-71)(n-72)(3n^2-24n+47)a(n-73)+$
 $(n-71)(n-70)(n-72)(n-73)(3n^2-21n+35)a(n-74)+$
 $(n-72)(n-71)(n-73)(n-74)(3n^2-24n+47)a(n-75)+$
 $(n-73)(n-72)(n-74)(n-75)(3n^2-21n+35)a(n-76)+$
 $(n-74)(n-73)(n-75)(n-76)(3n^2-24n+47)a(n-77)+$
 $(n-75)(n-74)(n-76)(n-77)(3n^2-21n+35)a(n-78)+$
 $(n-76)(n-75)(n-77)(n-78)(3n^2-24n+47)a(n-79)+$
 $(n-77)(n-76)(n-78)(n-79)(3n^2-21n+35)a(n-80)+$
 $(n-78)(n-77)(n-79)(n-80)(3n^2-24n+47)a(n-81)+$
 $(n-79)(n-78)(n-80)(n-81)(3n^2-21n+35)a(n-82)+$
 $(n-80)(n-79)(n-81)(n-82)(3n^2-24n+47)a(n-83)+$
 $(n-81)(n-80)(n-82)(n-83)(3n^2-21n+35)a(n-84)+$
 $(n-82)(n-81)(n-83)(n-84)(3n^2-24n+47)a(n-85)+$
 $(n-83)(n-82)(n-84)(n-85)(3n^2-21n+35)a(n-86)+$
 $(n-84)(n-83)(n-85)(n-86)(3n^2-24n+47)a(n-87)+$
 $(n-85)(n-84)(n-86)(n-87)(3n^2-21n+35)a(n-88)+$
 $(n-86)(n-85)(n-87)(n-88)(3n^2-24n+47)a(n-89)+$
 $(n-87)(n-86)(n-88)(n-89)(3n^2-21n+35)a(n-90)+$
 $(n-88)(n-87)(n-89)(n-90)(3n^2-24n+47)a(n-91)+$
 $(n-89)(n-88)(n-90)(n-91)(3n^2-21n+35)a(n-92)+$
 $(n-90)(n-89)(n-91)(n-92)(3n^2-24n+47)a(n-93)+$
 $(n-91)(n-90)(n-92)(n-93)(3n^2-21n+35)a(n-94)+$
 $(n-92)(n-91)(n-93)(n-94)(3n^2-24n+47)a(n-95)+$
 $(n-93)(n-92)(n-94)(n-95)(3n^2-21n+35)a(n-96)+$
 $(n-94)(n-93)(n-95)(n-96)(3n^2-24n+47)a(n-97)+$
 $(n-95)(n-94)(n-96)(n-97)(3n^2-21n+35)a(n-98)+$
 $(n-96)(n-95)(n-97)(n-98)(3n^2-24n+47)a(n-99)+$
 $(n-97)(n-96)(n-98)(n-99)(3n^2-21n+35)a(n-100)+$
 $(n-98)(n-97)(n-99)(n-100)(3n^2-24n+47)a(n-101)+$
 $(n-99)(n-98)(n-100)(n-101)(3n^2-21n+35)a(n-102)+$
 $(n-100)(n-99)(n-101)(n-102)(3n^2-24n+47)a(n-103)+$
 $(n-101)(n-100)(n-102)(n-103)(3n^2-21n+35)a(n-104)+$
 $(n-102)(n-101)(n-103)(n-104)(3n^2-24n+47)a(n-105)+$
 $(n-103)(n-102)(n-104)(n-105)(3n^2-21n+35)a(n-106)+$
 $(n-104)(n-103)(n-105)(n-106)(3n^2-24n+47)a(n-107)+$
 $(n-105)(n-104)(n-106)(n-107)(3n^2-21n+35)a(n-108)+$
 $(n-106)(n-105)(n-107)(n-108)(3n^2-24n+47)a(n-109)+$
 $(n-107)(n-106)(n-108)(n-109)(3n^2-21n+35)a(n-110)+$
 $(n-108)(n-107)(n-109)(n-110)(3n^2-24n+47)a(n-111)+$
 $(n-109)(n-108)(n-110)(n-111)(3n^2-21n+35)a(n-112)+$
 $(n-110)(n-109)(n-111)(n-112)(3n^2-24n+47)a(n-113)+$
 $(n-111)(n-110)(n-112)(n-113)(3n^2-21n+35)a(n-114)+$
 $(n-112)(n-111)(n-113)(n-114)(3n^2-24n+47)a(n-115)+$
 $(n-113)(n-112)(n-114)(n-115)(3n^2-21n+35)a(n-116)+$
 $(n-114)(n-113)(n-115)(n-116)(3n^2-24n+47)a(n-117)+$
 $(n-115)(n-114)(n-116)(n-117)(3n^2-21n+35)a(n-118)+$
 $(n-116)(n-115)(n-117)(n-118)(3n^2-24n+47)a(n-119)+$
 $(n-117)(n-116)(n-118)(n-119)(3n^2-21n+35)a(n-120)+$
 $(n-118)(n-117)(n-119)(n-120)(3n^2-24n+47)a(n-121)+$
 $(n-119)(n-118)(n-120)(n-121)(3n^2-21n+35)a(n-122)+$
 $(n-120)(n-119)(n-121)(n-122)(3n^2-24n+47)a(n-123)+$
 $(n-121)(n-120)(n-122)(n-123)(3n^2-21n+35)a(n-124)+$
 $(n-122)(n-121)(n-123)(n-124)(3n^2-24n+47)a(n-125)+$
 $(n-123)(n-122)(n-124)(n-125)(3n^2-21n+35)a(n-126)+$
 $(n-124)(n-123)(n-125)(n-126)(3n^2-24n+47)a(n-127)+$
 $(n-125)(n-124)(n-126)(n-127)(3n^2-21n+35)a(n-128)+$
 $(n-126)(n-125)(n-127)(n-128)(3n^2-24n+47)a(n-129)+$
 $(n-127)(n-126)(n-128)(n-129)(3n^2-21n+35)a(n-130)+$
 $(n-128)(n-127)(n-129)(n-130)(3n^2-24n+47)a(n-131)+$
 $(n-129)(n-128)(n-130)(n-131)(3n^2-21n+35)a(n-132)+$
 $(n-130)(n-129)(n-131)(n-132)(3n^2-24n+47)a(n-133)+$
 $(n-131)(n-130)(n-132)(n-133)(3n^2-21n+35)a(n-134)+$
 $(n-132)(n-131)(n-133)(n-134)(3n^2-24n+47)a(n-135)+$
 $(n-133)(n-132)(n-134)(n-135)(3n^2-21n+35)a(n-136)+$
 $(n-134)(n-133)(n-135)(n-136)(3n^2-24n+47)a(n-137)+$
 $(n-135)(n-134)(n-136)(n-137)(3n^2-21n+35)a(n-138)+$
 $(n-136)(n-135)(n-137)(n-138)(3n^2-24n+47)a(n-139)+$
 $(n-137)(n-136)(n-138)(n-139)(3n^2-21n+35)a(n-140)+$
 $(n-138)(n-137)(n-139)(n-140)(3n^2-24n+47)a(n-141)+$
 $(n-139)(n-138)(n-140)(n-141)(3n^2-21n+35)a(n-142)+$
 $(n-140)(n-139)(n-141)(n-142)(3n^2-24n+47)a(n-143)+$
 $(n-141)(n-140)(n-142)(n-143)(3n^2-21n+35)a(n-144)+$
 $(n-142)(n-141)(n-143)(n-144)(3n^2-24n+47)a(n-145)+$
 $(n-143)(n-142)(n-144)(n-145)(3n^2-21n+35)a(n-146)+$
 $(n-144)(n-143)(n-145)(n-146)(3n^2-24n+47)a(n-147)+$
 $(n-145)(n-144)(n-146)(n-147)(3n^2-21n+35)a(n-148)+$
 $(n-146)(n-145)(n-147)(n-148)(3n^2-24n+47)a(n-149)+$
 $(n-147)(n-146)(n-148)(n-149)(3n^2-21n+35)a(n-150)+$
 $(n-148)(n-147)(n-149)(n-150)(3n^2-24n+47)a(n-151)+$
 $(n-149)(n-148)(n-150)(n-151)(3n^2-21n+35)a(n-152)+$
 $(n-150)(n-149)(n-151)(n-152)(3n^2-24n+47)a(n-153)+$
 $(n-151)(n-150)(n-152)(n-153)(3n^2-21n+35)a(n-154)+$
 $(n-152)(n-151)(n-153)(n-154)(3n^2-24n+47)a(n-155)+$
 $(n-153)(n-152)(n-154)(n-155)(3n^2-21n+35)a(n-156)+$
 $(n-154)(n-153)(n-155)(n-156)(3n^2-24n+47)a(n-157)+$
 $(n-155)(n-154)(n-156)(n-157)(3n^2-21n+35)a(n-158)+$
 $(n-156)(n-155)(n-157)(n-158)(3n^2-24n+47)a(n-159)+$
 $(n-157)(n-156)(n-158)(n-159)(3n^2-21n+35)a(n-160)+$
 $(n-158)(n-157)(n-159)(n-160)(3n^2-24n+47)a(n-161)+$
 $(n-159)(n-158)(n-160)(n-161)(3n^2-21n+35)a(n-162)+$
 $(n-160)(n-159)(n-161)(n-162)(3n^2-24n+47)a(n-163)+$
 $(n-161)(n-160)(n-162)(n-163)(3n^2-21n+35)a(n-164)+$
 $(n-162)(n-161)(n-163)(n-164)(3n^2-24n+47)a(n-165)+$
 $(n-163)(n-162)(n-164)(n-165)(3n^2-21n+35)a(n-166)+$
 $(n-164)(n-163)(n-165)(n-166)(3n^2-24n+47)a(n-167)+$
 $(n-165)(n-164)(n-166)(n-167)(3n^2-21n+35)a(n-168)+$
 $(n-166)(n-165)(n-167)(n-168)(3n^2-24n+47)a(n-169)+$
 $(n-167)(n-166)(n-168)(n-169)(3n^2-21n+35)a(n-170)+$
 $(n-168)(n-167)(n-169)(n-170)(3n^2-24n+47)a(n-171)+$
 $(n-169)(n-168)(n-170)(n-171)(3n^2-21n+35)a(n-172)+$
 $(n-170)(n-169)(n-171)(n-172)(3n^2-24n+47)a(n-173)+$
 $(n-171)(n-170)(n-172)(n-173)(3n^2-21n+35)a(n-174)+$
 $(n-172)(n-171)(n-173)(n-174)(3n^2-24n+47)a(n-175)+$
 $(n-173)(n-172)(n-174)(n-175)(3n^2-21n+35)a(n-176)+$
 $(n-174)(n-173)(n-175)(n-176)(3n^2-24n+47)a(n-177)+$
 $(n-175)(n-174)(n-176)(n-177)(3n^2-21n+35)a(n-178)+$
 $(n-176)(n-175)(n-177)(n-178)(3n^2-24n+47)a(n-179)+$
 $(n-177)(n-176)(n-178)(n-179)(3n^2-21n+35)a(n-180)+$
 $(n-178)(n-177)(n-179)(n-180)(3n^2-24n+47)a(n-181)+$
 $(n-179)(n-178)(n-180)(n-181)(3n^2-21n+35)a(n-182)+$
 $(n-180)(n-179)(n-181)(n-182)(3n^2-24n+47)a(n-183)+$
 $(n-181)(n-180)(n-182)(n-183)(3n^2-21n+35)a(n-184)+$
 $(n-182)(n-181)(n-183)(n-184)(3n^2-24n+47)a(n-185)+$
 $(n-183)(n-182)(n-184)(n-185)(3n^2-21n+35)a(n-186)+$
 $(n-184)(n-183)(n-185)(n-186)(3n^2-24n+47)a(n-187)+$
 $(n-185)(n-184)(n-186)(n-187)(3n^2-21n+35)a(n-188)+$
 $(n-186)(n-185)(n-187)(n-188)(3n^2-24n+47)a(n-189)+$
 $(n-187)(n-186)(n-188)(n-189)(3n^2-21n+35)a(n-190)+$
 $(n-188)(n-187)(n-189)(n-190)(3n^2-24n+47)a(n-191)+$
 $(n-189)(n-188)(n-190)(n-191)(3n^2-21n+35)a(n-192)+$
 $(n-190)(n-189)(n-191)(n-192)(3n^2-24n+47)a(n-193)+$
 $(n-191)(n-190)(n-192)(n-193)(3n^2-21n+35)a(n-194)+$
 $(n-192)(n-191)(n-193)(n-194)(3n^2-24n+47)a(n-195)+$
 $(n-193)(n-192)(n-194)(n-195)(3n^2-21n+35)a(n-196)+$
 $(n-194)(n-193)(n-195)(n-196)(3n^2-24n+47)a(n-197)+$
 $(n-195)(n-194)(n-196)(n-197)(3n^2-21n+35)a(n-198)+$
 $(n-196)(n-195)(n-197)(n-198)(3n^2-24n+47)a(n-199)+$
 $(n-197)(n-196)(n-198)(n-199)(3n^2-21n+35)a(n-200)+$
 $(n-198)(n-197)(n-199)(n-200)(3n^2-24n+47)a(n-201)+$
 $(n-199)(n-198)(n-200)(n-201)(3n^2-21n+35)a(n-202)+$
 $(n-200)(n-199)(n-201)(n-202)(3n^2-24n+47)a(n-203)+$
 $(n-201)(n-200)(n-202)(n-203)(3n^2-21n+35)a(n-204)+$
 $(n-202)(n-201)(n-203)(n-204)(3n^2-24n+47)a(n-205)+$
 $(n-203)(n-202)(n-204)(n-205)(3n^2-21n+35)a(n-206)+$
 $(n-204)(n-203)(n-205)(n-206)(3n^2-24n+47)a(n-207)+$
 $(n-205)(n-204)(n-206)(n-207)(3n^2-21n+35)a(n-208)+$
 $(n-206)(n-205)(n-207)(n-208)(3n^2-24n+47)a(n-209)+$
 $(n-207)(n-206)(n-208)(n-209)(3n^2-21n+35)a(n-210)+$
 $(n-208)(n-207)(n-209)(n-210)(3n^2-24n+47)a(n-211)+$
 $(n-209)(n-208)(n-210)(n-211)(3n^2-21n+35)a(n-212)+$
 $(n-210)(n-209)(n-211)(n-212)(3n^2-24n+47)a(n-213)+$
 $(n-211)(n-210)(n-212)(n-213)(3n^2-21n+35)a(n-214)+$
 $(n-212)(n-211)(n-213)(n-214)(3n^2-24n+47)a(n-215)+$
 $(n-213)(n-212)(n-214)(n-215)(3n^2-21n+35)a(n-216)+$
 $(n-214)(n-213)(n-215)(n-216)(3n^2-24n+47)a(n-217)+$
 $(n-215)(n-214)(n-216)(n-217)(3n^2-21n+35)a(n-218)+$
 $(n-216)(n-215)(n-217)(n-218)(3n^2-24n+47)a(n-219)+$
 $(n-217)(n-216)(n-218)(n-219)(3n^2-21n+35)a(n-220)+$
 $(n-218)(n-217)(n-219)(n-220)(3n^2-24n+47)a(n-221)+$
 $(n-219)(n-218)(n-220)(n-221)(3n^2-21n+35)a(n-222)+$
 $(n-220)(n-219)(n-221)(n-222)(3n^2-24n+47)a(n-223)+$
 $(n-221)(n-220)(n-222)(n-223)(3n^2-21n+35)a(n-224)+$
 $(n-222)(n-221)(n-223)(n-224)(3n^2-24n+47)a(n-225)+$
 $(n-223)(n-222)(n-224)(n-225)(3n^2-21n+35)a(n-226)+$
 $(n-224)(n-223)(n-225)(n-226)(3n^2-24n+47)a(n-227)+$
 $(n-225)(n-224)(n-226)(n-227)(3n^2-21n+35)a(n-228)+$
 $(n-226)(n-225)(n-227)(n-228)(3n^2-24n+47)a(n-229)+$
 $(n-227)(n-226)(n-228)(n-229)(3n^2-21n+35)a(n-230)+$
 $(n-228)(n-227)(n-229)(n-230)(3n^2-24n+47)a(n-231)+$
 $(n-229)(n-228)(n-230)(n-231)(3n^2-21n+35)a(n-232)+$
 $(n-230)(n-229)(n-231)(n-232)(3n^2-24n+47)a(n-233)+$
 $(n-231)(n-230)(n-232)(n-233)(3n^2-21n+35)a(n-234)+$
 $(n-232)(n-231)(n-233)(n-234)(3n^2-24n+47)a(n-235)+$
 $(n-233)(n-232)(n-234)(n-235)(3n^2-21n+35)a(n-236)+$
 $(n-234)(n-233)(n-235)(n-236)(3n^2-24n+47)a(n-237)+$
 $(n-235)(n-234)(n-236)(n-237)(3n^2-21n+35)a(n-238)+$
 $(n-236)(n-235)(n-237)(n-238)(3n^2-24n+47)a(n-239)+$
 $(n-237)(n-236)(n-238)(n-239)(3n^2-21n+35)a(n-240)+$
 $(n-238)(n-237)(n-239)(n-240)(3n^2-24n+47)a(n-241)+$
 $(n-239)(n-238)(n-240)(n-241)(3n^2-21n+35)a(n-242)+$
 $(n-240)(n-239)(n-241)(n-242)(3n^2-24n+47)a(n-243)+$
 $(n-241)(n-240)(n-242)(n-243)(3n^2-21n+35)a(n-244)+$
 $(n-242)(n-241)(n-243)(n-244)(3n^2-24n+47)a(n-245)+$
 $(n-243)(n-242)(n-244)(n-245)(3n^2-21n+35)a(n-246)+$
 $(n-244)(n-243)(n-245)(n-246)(3n^2-24n+47)a(n-247)+$
 $(n-245)(n-244)(n-246)(n-247)(3n^2-21n+35)a(n-248)+$
 $(n-246)(n-245)(n-247)(n-248)(3n^2-24n+47)a(n-249)+$
 $(n-247)(n-246)(n-248)(n-249)(3n^2-21n+35)a(n-250)+$
 $(n-248)(n-247)(n-249)(n-250)(3n^2-24n+47)a(n-251)+$
 $(n-249)(n-248)(n-250)(n-251)(3n^2-21n+35)a(n-252)+$
 $(n-250)(n-249)(n-251)(n-252)(3n^2-24n+47)a(n-253)+$
 $(n-251)(n-2$